

The idea is to check the time to complete a rotation today and again after a century. If we suppose that it takes exactly 24 hours for Earth to complete one revolution today, this time would have reduced by $1.30 \times 10^{-3} \text{ s}$, a century from now.

(a) We will first determine how many seconds are there in one century.

$$1 \text{ Century} = 100 \text{ yr} \left(\frac{365 \text{ day}}{1 \text{ yr}} \right) \left(\frac{24 \text{ hr}}{1 \text{ day}} \right) \left(\frac{60 \text{ min}}{1 \text{ hr}} \right) \left(\frac{60 \text{ s}}{1 \text{ min}} \right)$$

No full conversion, No point

$$= 3.1536 \times 10^9 \text{ s in 1 century.}$$

The angular acceleration is determined by examining the angular velocities of Earth today and after a century.

$$\therefore \omega_{\text{today}} = \frac{\Delta \theta}{\Delta t} = \frac{2\pi \text{ rad}}{24 \text{ hr} \times \frac{3600 \text{ s}}{1 \text{ hr}}} = 7.272205216643 \times 10^{-5} \text{ rad/s}$$

$$\text{Also, } \omega_{\text{after century}} = \frac{\Delta \theta}{\Delta t} = \frac{2\pi \text{ rad}}{\left[(24 \text{ hr} \times \frac{3600 \text{ s}}{1 \text{ hr}}) - 1.30 \times 10^{-3} \text{ s} \right]}$$

$$= 7.27220532606 \times 10^{-5} \text{ rad/s}$$

$$\therefore \alpha = \frac{\Delta \omega}{\Delta t} \Big|_{\text{century}} = \frac{-1.0941975606344 \times 10^{-12} \text{ rad/s}}{3.1536 \times 10^9 \text{ s}} \quad \text{if } \Delta \omega = \omega_{\text{today}} - \omega_{\text{cent.}}$$

$$= -3.47 \times 10^{-22} \text{ rad/s}^2$$

(+) or (-) sign ok!

Units, sig. figs

NB

Do not round up during the intermediate steps !!!

[13] Question Total.

(8)

(b) Determining the torque exerted by the moon on planet Earth.

$$\tau = I\alpha$$

$$= \left(\frac{2}{5}MR^2\right)\alpha$$

$$= \frac{2}{5}(5.974 \times 10^{24} \text{ kg})(6.378 \times 10^6 \text{ m})\alpha$$

with $\alpha = -3.469 \times 10^{-22} \text{ rad/s}^2$.

$$= -3.3727 \times 10^{16} \text{ N}\cdot\text{m}$$
$$(-3.37 \times 10^{16} \text{ N}\cdot\text{m})$$

Look up I for a solid sphere - Table 10.2.

And from Appendix:

$$M_E = 5.974 \times 10^{24} \text{ kg}$$

$$R_E = 6.378 \times 10^6 \text{ m}$$

(c) $\vec{\tau} = \vec{r} \times \vec{F}$ angle between $\vec{r}$ and $\vec{F} = 90^\circ$ Extra point
 $\Rightarrow \tau = r_{E\text{-moon}} F_{\text{man}}$ If they started off at the definition of torque.

$$\therefore r_{E\text{-moon}} = \frac{\tau}{F_{\text{man}}} = \frac{3.3727 \times 10^{16} \text{ N}\cdot\text{m}}{850 \text{ N}}$$

$$= 3.9679 \times 10^{13} \text{ m}$$

$$(3.98 \times 10^{13} \text{ m})$$

(2)

The curved path that Jane would take, is approximately equal to D , the width of the river. It can be expressed in terms of the angles θ and ϕ as:

$$D = L \sin \phi + L \sin \theta$$

Both θ and L are known, D too

$\Rightarrow$ We can solve for ϕ .

$$L \sin \phi = D - L \sin \theta$$

$$\sin \phi = \frac{D - L \sin \theta}{L}$$

$$\Rightarrow \phi = \sin^{-1} \left(\frac{D}{L} - \sin \theta \right) = \sin^{-1} \left(\frac{51.0 \text{ m}}{40.0 \text{ m}} - \sin 50.0^\circ \right)$$

$$= 30.6^\circ \checkmark$$

If we choose Jane + Boat to be our system, we can write that

$$\Delta K + \Delta U_g = W, \text{ where } W \text{ is the work done by the wind.}$$

With the surfer being approximately the river width, $\vec{F}$ and $d\vec{l}$ will be in opposite directions, in the work done formulation: $W = \int \vec{F} \cdot d\vec{l} = -FD \checkmark$

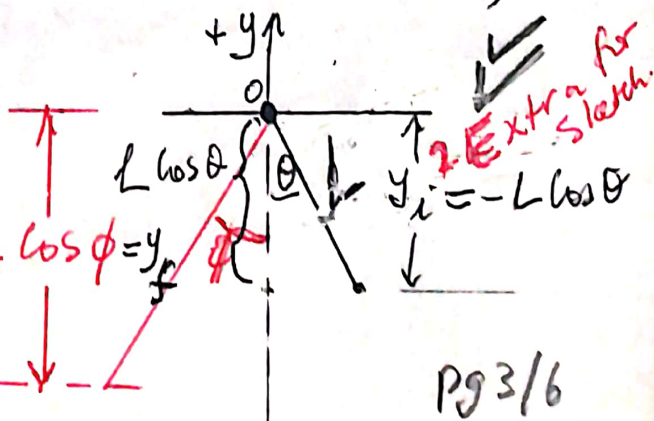
We will let Jane have an initial velocity; but her velocity at the other bank equal to 0. $\checkmark$

We can anchor the potential at the point where the rope attaches to the tree. As such,

$$\left. \begin{aligned} y_i &= -L \cos \theta \\ \text{and} \\ y_f &= -L \cos \phi \end{aligned} \right\} \begin{aligned} &\text{measured from} \\ &\text{point where rope} \\ &\text{attaches to tree} \end{aligned}$$

$$\therefore \left(\frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2 \right) + (m g y_f - m g y_i) = -FD$$

$$0 - \frac{1}{2} m v_i^2 - m g L \cos \phi + m g L \cos \theta = -FD$$



$$\frac{1}{2}mv_i^2 + mLg \cos \phi - mLg \cos \theta = FD$$

$$\therefore v_i^2 = \frac{2}{m} [FD + mLg \cos \theta - mLg \cos \phi]$$

$$v_i = \sqrt{\frac{2FD}{m} + 2Lg (\cos \theta - \cos \phi)}$$

✓ analytic expression

$$= \sqrt{\frac{2(111\text{N})(51.0\text{m})}{50.0\text{kg}} + 2(40.0\text{m})(9.80\text{m/s}^2) \times}$$

$$\sqrt{\cos 50.0^\circ - \cos 30.6^\circ}$$

$$= 7.45\text{ m/s} \checkmark$$

(10)

(b) For swinging back:

FD is positive but the energy equation is still the same.

$$\Delta K + \Delta U_g = W \checkmark$$

$$\text{let } M = m_{\text{Jane}} + m_{\text{Torran}} = 130.0\text{kg} \checkmark$$

$$(0 - \frac{1}{2}Mv_i^2) + (Mgy_f - Mgy_i) = FD$$

$$-\frac{1}{2}Mv_i^2 + Mg(-L \cos \theta) - Mg(-L \cos \phi) = FD$$

$$\therefore v_i = \sqrt{2gL(\cos \phi - \cos \theta) - \frac{2FD}{M}}$$

$$= \sqrt{2(9.80\text{m/s}^2)(40.0\text{m})[\cos 50.0^\circ - \cos 30.6^\circ]}$$

$$\sqrt{\frac{2(111\text{N})(50.1\text{m})}{130.0\text{kg}}}$$

$$= 9.16\text{ m/s} \checkmark$$

(4)

Total [14]

$$V_T = \sqrt{\frac{2mg}{D\rho_{\text{air}}A}}$$

[10]

$$r_{\text{drop}} = 2.00 \times 10^{-3} \text{ m}$$

$$h = 1000 \text{ m}$$

$$D = 0.50$$

$$= \sqrt{\frac{2(\rho_{\text{water}} V_{\text{mass}})g}{D\rho_{\text{air}}\pi r_{\text{drop}}^2}}$$

Conversion for mass of droplet.

look up parameters/constants

$$\rho_{\text{water}} = 1000 \text{ kg/m}^3$$

$$\rho_{\text{air}} = 1.20 \text{ kg/m}^3 @ 20^\circ\text{C}$$

$$= \sqrt{\frac{2\rho_{\text{water}} \frac{4}{3}\pi r_{\text{drop}}^3 \cdot g}{D\rho_{\text{air}}\pi r_{\text{drop}}^2}}$$

Cross-sectional area

$$= \sqrt{\frac{2\rho_{\text{water}}(\frac{4}{3})r_{\text{drop}} \cdot g}{D\rho_{\text{air}}\pi}} = 9.332 \text{ m/s}$$

Correct substitution

$$\left\{ \frac{2(1000 \text{ kg/m}^3)(\frac{4}{3})(2.00 \times 10^{-3} \text{ m})(9.80 \text{ m/s}^2)}{(0.5)(1.20 \text{ kg/m}^3)} \right\}$$

← post simplification!

(b) No, Terminal speed is attained when the particle experiences zero net acceleration. Therefore, the drop released from 1000m has not attained this condition 'yet' when it hits the ground. There is no discrepancy. (3)

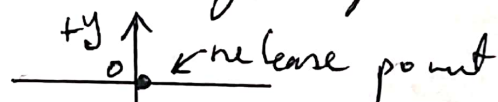
(c) Calculating the speed of the raindrop after its release from 1000m and when it is nearly at ground.

$$v_{yf}^2 = v_{yi}^2 + 2a_y(y_f - y_i)$$

$$= 0 + 2(-9.80 \text{ m/s}^2)(-1000 \text{ m} - 0)$$

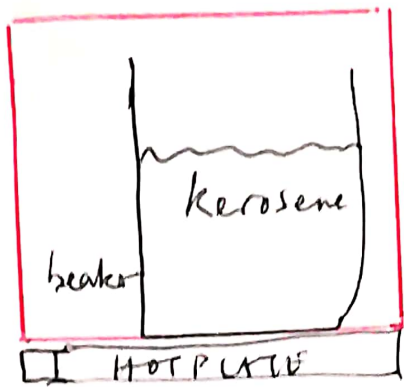
$$= 140 \text{ m/s}$$

$$\text{But } 140 \text{ m/s} = \frac{140 \times 10^{-3} \text{ km}}{1 \text{ s} (\frac{1 \text{ hr}}{3600 \text{ s}})} = 504 \text{ km/h}$$



Extra for neat & complete sketch!!

-1000 m (at ground)



[6]

$$m_b = 300 \text{ g} = 0.300 \text{ kg}$$

$$c_b = 490 \text{ J/(kg} \cdot ^\circ\text{C)}$$

$$m_{\text{kerosene}} = 900 \text{ g} = 0.900 \text{ kg}$$

$$c_{\text{kerosene}} = 2018 \text{ J/(kg} \cdot ^\circ\text{C)}$$

$$\frac{\Delta T}{\Delta t} = 2.10^\circ\text{C per minute}$$

Determine $\frac{Q}{\Delta t}$ which is power.

$$Q_{\text{cold}} = -Q_{\text{hot}} \text{ or by source}$$

$$\therefore -Q_{\text{hot}} = Q_{\text{beaker}} + Q_{\text{kerosene}} \checkmark$$

$$= m_b c_b \Delta T + m_{\text{kero}} c_{\text{kero}} \Delta T \checkmark$$

$$= (m_b c_b + m_{\text{kero}} c_{\text{kero}}) \Delta T$$

$$\text{But } P = \frac{Q_{\text{source}}}{\Delta t} = (m_b c_b + m_{\text{kero}} c_{\text{kero}}) \frac{\Delta T}{\Delta t} \checkmark$$

$$= [(0.300 \text{ kg})(490 \text{ J/(kg} \cdot ^\circ\text{C)})] + [(0.900 \text{ kg})(2018 \text{ J/kg} \cdot ^\circ\text{C})]$$

$$\times \left(\frac{\Delta T}{\Delta t} = \frac{2.10^\circ\text{C}}{60 \text{ sec}} \right)$$

$$= \frac{4122.72 \text{ J}}{60 \text{ sec}} \checkmark$$

$$= 68.712 \text{ W}$$

$$(68.7 \text{ W}) \checkmark$$

6